



Team Details

Team Name:

Web Shooters

SR. NO	ROLE	NAME	ACADEMIC YEAR
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Problem Statement Addressed



KLA PS01 — AI-Based Restoration of Degraded Semiconductor Inspection Images

DESCRIPTION / DETAILS

- Inspection images decide whether a wafer passes: one pixel of noise can hide a killer defect.
- Three degradations, order undisclosed — speckle noise, additive Gaussian noise, 2× downsampling.
- Input 128×128 noisy → output 256×256 clean (also 256→512). Grayscale float32.
- Degraded pixels fall outside [0,1] by design; ground truth always inside it.
- Scored on three axes: PSNR + SSIM + LPIPS, H100 end-to-end throughput, training/compute hygiene.
- Hidden test set contains out-of-distribution content — must generalise, not memorise.



Idea Description – Describe your Idea/Solution/Prototype



One compact fully-convolutional network denoises and 2× super-resolves in a single pass, built on a degradation model recovered from the data rather than guessed.

KEY CONCEPT & APPROACH

- Reverse-engineered KLA's exact degradation recipe from the 3200 provided pairs.
- That yields unlimited correctly-degraded training data, with fresh noise every epoch.

SOLUTION OVERVIEW

- 4.4M-parameter residual CNN; all work at low resolution, one PixelShuffle at the end – 4× cheaper.
- Fully convolutional, so the same weights restore 128→256 and 256→512.



Proposed Solution – Describe your Idea/Solution/Prototype



Measured forward model → unlimited synthetic pairs → compact CNN → loss blended for pixel and perceptual quality.

SOLUTION DETAILS

- DEGRADATION RECOVERED (measured): 2×2 area-average downsample; noise applied after it; speckle multiplicative, $\text{var} \propto \text{pixel}^2$ ($r = 0.993$); $\sigma_{\text{speckle}} 0.10\text{--}0.25$, $\sigma_{\text{gauss}} 0.00\text{--}0.15$.
- verify_degradation.py re-derives every constant from the data and ABORTS on mismatch.
- TRAINING: 50/50 real and synthesised pairs; noise ranges widened beyond measured; 8-fold dihedral augmentation.
- MODEL: 24 residual blocks $\times$ 96 ch $\rightarrow$ PixelShuffle $\times 2$. No global skip from the noisy input. Output clamped to $[0,1]$.
- LOSS: Charbonnier (smooth L1) for pixel accuracy, then a fine-tune adding a frozen-AlexNet LPIPS term for perceptual quality.
- STACK: PyTorch 2.9.1+cu128 — large_ft_perceptual 25 ep/3.5 h + large 147 ep/6.1 h + large_perc_long 40 ep/6.7 h on RTX 4050 (base) and GTX 1650 (fine-tune).
- FEASIBILITY: 50 MB checkpoint, 101 ms/image end-to-end on a GTX 1650; runs unchanged on CPU or any NVIDIA GPU.



Innovation and Uniqueness



We measured what others assume — which produced both the data advantage and the model.

KEY INNOVATION

- Degradation recovered, not guessed — giving unlimited perfectly-matched training data.
- Noise SPECTRUM matched, not just variance: caught a 16% high-frequency error every variance test passed.
- Perceptual fine-tune of a pixel-trained network: LPIPS 0.3099 → 0.1811, a 42% cut, for 0.34 dB of PSNR. The interpolation curve between the two was measured and showed no blend was needed.

COMPETITIVE ADVANTAGE

- Beats bicubic on 200/200 validation images — every one, not on average.
- Compact by design (4.4M params, 50 MB): all computation at low resolution, so throughput stays high.
- Three ideas tested and REJECTED on evidence: 8× TTA (worse LPIPS, 3 checkpoints), synthetic-pattern training (−0.22 dB), longer perceptual schedule (won the val split, lost 0.57 dB out-of-distribution).



Impact and Benefits



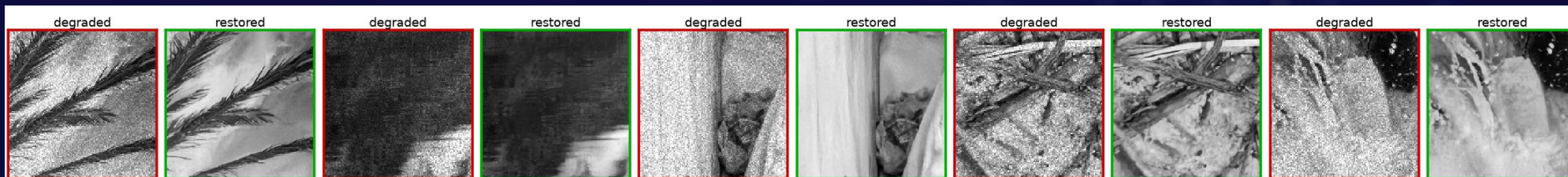
Recovers detail engineers currently accept as lost — and generalises to structure it never trained on.

Primary Impact

- PSNR 23.23 → 28.68 dB
- SSIM 0.5395 → 0.7543
- LPIPS 0.4369 → 0.1811
- Untrained control 10.35 dB — the metric can fail.

Quantifiable Outcomes

- +5.44 dB PSNR, +0.2148 SSIM, 0.2558 better LPIPS
- 101 ms/image end-to-end (read → preprocess → transfer → model → save)
- Unseen semiconductor-like structure: +2.32 dB mean, ahead on 13/15 patterns





GitHub & Video Link



GitHub Repository

https://github.com/bevinj2255/i4c_hackathon



Prototype / Simulation Video

Not submitted — full results, figures and code are in the repository.

REFERENCES

Lim et al., EDSR, CVPRW 2017 · Shi et al., PixelShuffle, CVPR 2016 · Zhang et al., LPIPS, CVPR 2018 · Zhai et al., IEEE Access 11:21049, 2023 · Terven et al., Artif Intell Rev 58:195, 2025 · Monga et al., IEEE SPM 38(2), 2021.

No external datasets or pretrained weights were used for the model.